

The ModelSEED Biochemistry Database, 2026 update: grading multi-source thermodynamics

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Abstract

The ModelSEED Biochemistry Database (<https://modelseed.org>) supplies foundational mass- and charge-balanced reaction networks for metabolic reconstructions. Here we present an update on the biochemistry where the database has been expanded to ~ 46,000 compounds, and ~ 48,000 reactions, featuring ~ 37,000 metabolic structures. We have expanded our approach for handling thermodynamic data, enabling multiple sources of data to be derived, integrated, and presented to the wider research community. We now publish predictions of pKa, reaction energy (and respective uncertainties), and estimates of reaction direction from multiple independent sources. Each reaction is graded gold, silver or bronze according to the strength of the evidence behind it, so that users can weigh its reliability directly. We also release reaction directions predicted by an ensemble of large language models. This multi-source approach exposes agreements and discrepancies between sources for ~ 27,000 reactions. Finally, to ensure data integrity, a new conflict-resolution pipeline reconciles structures across sources, documenting input from curators. Our work is publicly available at <https://github.com/ModelSEED/ModelSEEDDatabase>.

Keywords biochemistry database, metabolic modelling, thermodynamics, reaction reversibility, ModelSEED

Introduction

The ModelSEED Biochemistry Database provides mass- and charge-balanced reaction networks reconciled across major biochemical resources for metabolic reconstructions (1). Since its 2020 release, the database has been widely adopted for genome-scale metabolic modeling and integration. For example, GEMsembler utilizes the shared ModelSEED namespace to resolve cross-tool identifier mappings (2), and Yeast9 incorporates its metabolite values to parameterize reaction Gibbs energies across the network (3).

This update reports the growth of the database, and an expanded approach in handling the layer of thermodynamics data allowing us to publish every source with its own uncertainty rather than promoting one. While over 70% of the reactions with multiple sources of data agree on reaction direction, any disagreement is published for review by researchers to decide which numbers and approach carry weight.

Here we also describe a grading approach incorporating the uncertainties of the data across different sources, an integration

of atom-mapping data to enable MFA analyses, and a structure-curation pipeline built to resolve conflict between structures from different databases. Finally, we expanded our search interface at <https://modelseed.org> for dissemination of the results.

Materials and Methods

Construction and Curation

We collated additional compounds and reactions from MetaCyc (4), Rhea (5), and ChEBI (6). The compounds are reconciled into single ModelSEED compounds using available structures where possible and the reactions are reconciled based on the identity of the compounds. The procedure follows the 2020 release (1). Mass and charge balance is enforced and flagged.

For the 2020 release, a conflict between structures from multiple sources was either manually reviewed or dropped silently. The updated pipeline classifies each structural conflict and routes it to one of three outcomes: an automatic pick from the source variant matching the historical formula, a curator-authored override with recorded rationale, or an entry in a mass-balance exclusion registry for compounds where no source formula is defensible. Every outcome is attributed to its curator and its evidence. We retained every structure we collect, so conflict resolution can be inspected within the data in the repository. An outline of how a researcher can contribute is given in Supplementary Methods S1 and expanded in the repository.

Multi-source thermodynamics

The 2020 release combined two sources: the historical group contribution approach (7) and eQuilibrator (8). This update carries four sources, each stored as its own record: group contribution, eQuilibrator 3.0 (9), dGPredictor (10), and experimental values where they exist. Only the experimental values carry a published direction. The three predictors were released as energy estimators, so every direction reported here is an inference from their energies. All values are reported at pH 7.0, ionic strength 0.25 M, pMg 3.0 and 298.15 K; the basis for each condition and its effect on the released energies are given in Supplementary Methods S2. ~~A key caveat: as we publish a generalized database that can be applied across the kingdoms of life, transport reactions are scored from stoichiometry and energy alone, and neither an estimate of membrane potential or pH gradients are used.~~

Protonation.

eQuilibrator's Legendre transform requires a *macroscopic* pKa ladder per compound: an ordered list in which consecutive species differ by one proton. The count of entries around the desired pH sets the proton count of the major microspecies, so the composition of the ladder changes the transformed energy directly. We generate the pKa ladders using ChemAxon Marvin (11) for use with eQuilibrator, but we're conscious that these data cannot be regenerated without a license, so we also include the pKas generated by an open-source alternative, MolGpKa (12), for every compound with a structure, and the

measured ladders of Alberty (13), versioned per structure source in an extended pKa dictionary.

Experimental anchors.

eQuilibrator and dGPredictor draw on the same set of experimental values derived from the NIST Thermodynamics of Enzyme-catalyzed Reactions (TECR) Database (14). ~~The work was keyed to KEGG, and eQuilibrator uses eQuilibrator's MetaNetX mappings to integrate with other resources but there are inherent structural conflicts when we examine the mappings to other resources conflict structurally with ModelSEED identifiers. In order to draw direct comparisons between the results generated for any ModelSEED reactions we~~ We re-built the experimental anchors of TECR values, curating ~~any~~ conflicts to ensure a direct mapping between TECR values and ModelSEED reactions. ~~In addition we, and~~ evaluated openTECR, a communal re-curation of the same source ~~available at (https://github.com/opentecr, to see if we can), to~~ extend the set of anchors.

Uncertainty.

The three sources each adopt a different approach for estimating the uncertainty surrounding the predicted energy of reaction ~~and as such, so~~ the scale of uncertainty is not directly comparable (Figure 2C) ~~median values (kcal·mol⁻¹) are 0.63 for eQuilibrator, 10.41 for group contribution, and 17.01 for dGPredictor; the median values are reported with the results in Section 3.3.~~ Comparing against the experimentally-anchored reactions, we find that group contribution overstates its error and eQuilibrator understates it.

Direction.

For historical reasons, we use the same heuristics for deriving reaction direction from the Group contribution approach (7) unchanged from 2020, so that series stays comparable across releases. For eQuilibrator and dGPredictor we use one rule set built on the reversibility index from Noor *et al.* (15): a reaction is called irreversible when $|\ln \Gamma|$ exceeds $\ln(1000)$ by more than one propagated standard deviation (9), Γ being the fold-change of reactant concentration required to change reaction direction). However, we extend the work by Noor *et al.* to require the uncertainty to clear the threshold as well. We report fewer directions than the published index would on the same energies, and we separate *reversible* reactions from *undetermined* cases. An exception we implement is for reactions whose uncertainty is small relative to the threshold and yet still straddles it: meaning the estimate lies within one standard deviation of the threshold, and these we report as reversible.

Transport.

Compartments are relative indices — 0 inside, 1 outside — not named compartments, so one reaction can be instantiated against a bacterial membrane, a mitochondrion or a plastid envelope when a template builds a model. Because the database is not committed to a cell type, transport reactions are scored from stoichiometry and energy alone: neither membrane potential nor pH gradient enters the estimate. The consequence is specific. A primary active transporter carries ATP hydrolysis in its stoichiometry, every predictor estimates that tightly, and the reaction earns a confident grade from chemistry that was never in question — 96% of gold-graded transport reactions are

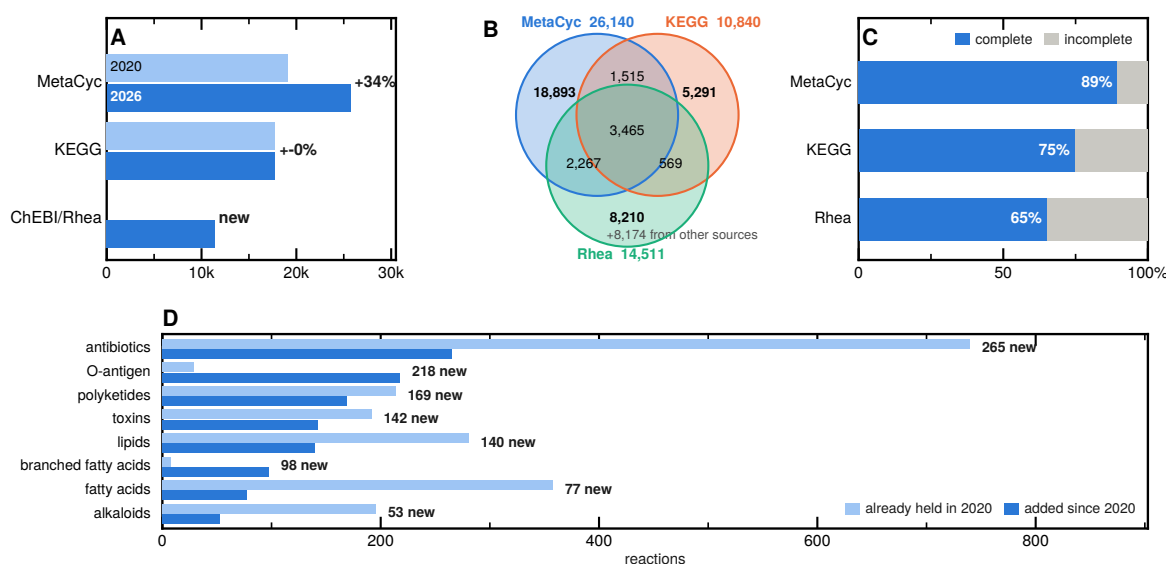


Figure 1 Sources of compounds and reactions. Sources of compounds and reactions, and where the growth landed. (A) compounds contributed by each source that also supplied molecular structures, 2020 against 2026: we did not introduce any new biochemistry from KEGG; The third row is ChEBI, which supplies structures but no reactions. (B) how the share of each three primary database's reactions that reaction sources overlap, every region labelled with its reaction count. Circle areas are unique; not proportional — three-set areas cannot be solved exactly — so the counts carry the quantities and the circles only carry membership. Rhea identifies its compounds through ChEBI. (C) the share of those each source's unique reactions that are complete, meaning every reagent in a reaction is assigned a complete structure, so the reaction can be balanced and decomposed. The three panels share one row per (D) reactions added since 2020 against those already held, by MetaCyc pathway class, for the eight classes that gained most. Both bars are built through the same join from the MetaCyc source table, so they are comparable; the third row organisational class Super-Pathways is ChEBI in (A) and Rhea in (B) and (C) excluded. Only additions with a MetaCyc pathway assignment can be placed — 2, because ChEBI supplies structures but no 282 of the 12,259; the 8,409 reactions and that arrived through Rhea supplies reactions but carry no structures pathway ontology and are absent from the panel rather than counted as unannotated. Rhea identifies its compounds through ChEBI, so its completeness growth is dependent on ChEBI structures concentrated in specialised metabolism and cell-envelope biosynthesis: 41 of the 12,259 reactions added fall in a central-carbon or energy-metabolism class.

ATP-coupled and only 2.7% translocate a proton. Such grades measure confidence in the coupled chemistry, not evidence that the transporter runs that way in a given organism; every reaction carries an is-transport field so that this can be accounted for downstream. Reactions whose only change is of compartment carry no net chemistry and none is graded gold. dGPredictor commits on 1.4% of transport reactions against 17.3% elsewhere, so transport grades also rest on fewer independent sources.

Grading reaction direction

We apply a classification approach with which we grade the predicted reaction direction based on the available evidence. Very often sources and heuristics disagree, as is the case here, and we qualify what the reaction direction would be, and how reliable the evidence is using several tiers for ease of interpretation: gold, silver, or bronze. Our evaluation splits two ways, on the confidence of a source's own claim, fitted as a probability against the experimental anchors, and what the other sources make of it, by a weighted comparison on the same scale. Grades, self-assessments, and cross-source verdicts are stored in the biochemistry database, and can be viewed in the UI. The process of grading reactions is described fully in Supplementary Methods S3.

Ensemble LLMs predictions

Thermodynamic assignment is bounded by estimate coverage: energies cannot be computed for an incomplete reaction in terms of structures. Furthermore, every reaction is a member of a pathway within a cell, and the overarching drive of the pathway, dynamically changing metabolite concentrations as downstream enzymes process reagents, means a reaction may be driven in a direction on a level that supersedes our evaluation (16; 17; 18), leaving 23,790 reactions with no direction from any predictor. We run a complementary approach using an ensemble, a "council" of large language models (LLMs) to interpret the reaction based on its name and stoichiometry alone. Three LLMs independently make their predictions, a fourth audits, and a fifth adjudicates. We provide the prompts in the repository, and describe the process and its limits in Supplementary Methods S4. We integrate these predictions transparently in our database, but do not include them in our grading of the thermodynamic evidence.

The ensemble directs 22,963 of the reactions no thermodynamic source will commit on, roughly doubling the fraction of the database that carries a direction. Its accuracy is not that of an energy calculation: scored against measurement it recovers the direction 77% of the time against 98% for eQuilibrator, and 85% of its calls are simply the direction the equation is written in, so its errors fall almost entirely on

reactions that run in reverse (Supplementary Methods S4). We therefore release these calls as a separate, clearly-labelled layer, excluded from the evidence grading: they are for reactions a reconstruction would otherwise leave reversible by default, and should be reviewed before use rather than treated as evidence. Neither approach models the pathway context that can drive a reaction against its own standard-state energy as metabolite concentrations shift downstream (16; 17; 18); that remains a property of a model rather than of a reference database.

Atom mapping

Atom mappings are generated from the workflow of Huß *et al.* (19) where they refine the per-reaction output of the Reaction Decoder Tool (20) into mappings that can be used to span a metabolic reconstruction. The approach also resolves chemically equivalent atoms into symmetry groups, such as the two oxygens of CO₂ and the six equivalent terminal oxygens of pyrophosphate, so the ambiguity is apparent. The output is formatted and stored in the database, and can also be visualized in the UI.

Distribution and community process

The database is distributed as versioned flat files and JSON from the public repository, queried programmatically through a SOLR endpoint, and browsed through the web interface at <https://modelseed.org>. Every merge runs continuous integration that rebuilds the database from source records and fails on any regression in balance, structure coverage or schema conformance, so a curation change that breaks an invariant cannot be merged. Researchers can contribute by adding and curating structures and reactions, the repository has a guide for doing so, and new data can be merged via pull requests.

Results

Growth and coverage

The database has grown 34% in compounds and 5534% in reactions since 2020, counting only non-obsolete entries at both time points, driven mainly by MetaCyc and the BioCyc family. Reaction growth is now split across three primary databases (Figure 1B): MetaCyc carries 31,805–26,140 reactions of which 21,258–18,893 are supplied by no other primary source, KEGG 14,066–10,840 of which 5,756–291 are unique, and the newly-integrated Rhea 17,477–14,511 of which 8,339–210 are unique: reactions the database would not otherwise hold. Completeness of reactions differs (Figure 1C). We classify a complete reaction where every reagent can be assigned a complete molecular structure: 90% of MetaCyc's and 7675% of KEGG's reactions against 6665% of Rhea's. Overall structure coverage rose from 28,120–087 to 36,943–compounds—907 compounds. The new biochemistry is not distributed like the old (Figure 1D): the MetaCyc classes that grew are those of specialised metabolism — antibiotic and polyketide biosynthesis, O-antigen and cell-envelope assembly, branched and acylated lipids, alkaloids — while central carbon and energy metabolism account for 41 of the 12,259 reactions added. Central metabolism was already saturated in 2020; what a 2026

reconstruction gains is the periphery. The comparison covers the MetaCyc-sourced additions; Rhea, which supplied 8,409 of the new reactions, has no pathway ontology to place them in.

Structure curation

We built and ran our curation pipeline on all structures that would impact the set of roughly 9,000 reactions used in the ModelSEED and PlantSEED reconstruction templates—, since an error in one of those propagates into every model built from them. This scope reflects a curation priority, not a statement about how much of the database is usable: the templates are discussed in Section 3.5. The conflict pipeline resolved every previously conflicting compound in three curation rounds. i) Instances where the conflict existed at the level of mobile hydrogens were resolved automatically using historical formula ii) instances where the structures were structurally different were resolved manually iii) instances where resolution was not straight-forward were entered in the mass-balance exclusion registry where we fix an arbitrarily chosen formula because no source formula was defensible.

Thermodynamics

The three predictors cover comparable fractions of the database and disagree substantially about how well they do it (Figure 2B). Uncertainty is reported on scales differing by more than an order of magnitude (Figure 2C): a median of 0.630.62 kcal mol⁻¹ for eEquilibrator against 10.4110.83 kcal mol⁻¹ for group contribution and 17.0117.57 kcal mol⁻¹ for dGPredictor.

Calibrating the reported errors against measurement.

We use the experimental anchors to evaluate the error estimates. For each source we collect one held-out prediction per anchored reaction, together with the uncertainty that source reports for it, and divide the residual against the measured $\Delta_r G'^{\circ}$ by that uncertainty. Measured this way, eEquilibrator has a root mean square of 8.06 and a median of 2.27, with only 28.7% of reactions within one reported standard deviation and 46.3% within two, so it understates its error. Group contribution runs the other way, with a median of 0.32 and 73.6% within one standard deviation, and dGPredictor is the best scaled of the three, with a root mean square of 1.15 and 87.9% within one. A researcher choosing by smallest reported error would systematically choose the source that understates it, which is the argument against promoting any single source. The grading in the next section calibrates each source's confidence against measurement rather than taking its uncertainty at face value.

Direction assignment and evidence grading

Each source now generates its own prediction for the direction each reaction proceeds, and the three sources differ more in what they will commit to than in what they say (Figure 3A). eEquilibrator resolves 84.397.4% of the reactions it covers, group contribution 42.959.7% and dGPredictor 41.140.1%. Across the database, 30,157–24,594 reactions receive a direction or a positive statement of reversibility from at least one source and 25,855–23,790 receive none.

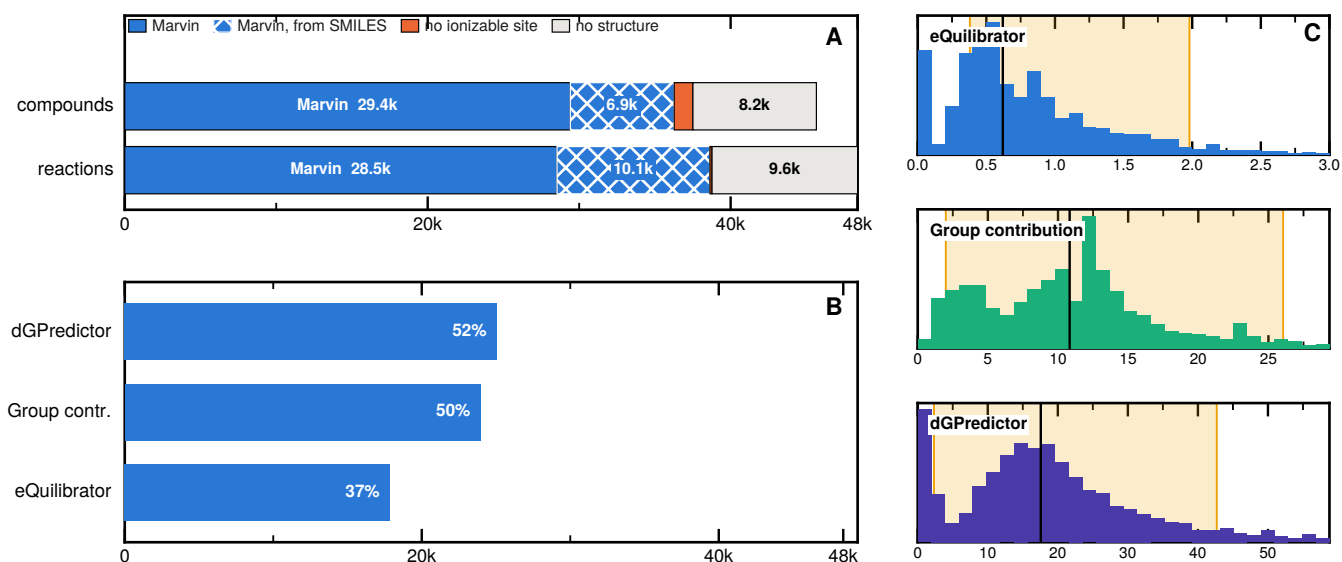


Figure 2 Thermodynamic sources. (A) Provenance of every pK_a ladder shipped in the database, across all 45,708,662 compounds and 5648,002 384 reactions. ChemAxon Marvin supplies 79.4% of compounds and 81.679.8% of reactions. Segments are identified by the key. Hatching marks the share reached through a second route: the calculator refuses polymers and organometallics as query molecules, so those are computed from SMILES instead. No ionizable site is a chemistry result — Marvin ran and reported no dissociable proton between pH -2 and 16, for 1,242 237 compounds — The remainder carry no structure to compute — and is coloured apart from no structure, which is a curation gap rather than a protonation one. (B) Reactions that carry an energy computed from the source heuristic. (C) Reported uncertainty in kcal mol⁻¹, one axis per source because the three differ by more than an order of magnitude. The shaded band spans the 5th to 95th percentile, which is used in our classification scheme.

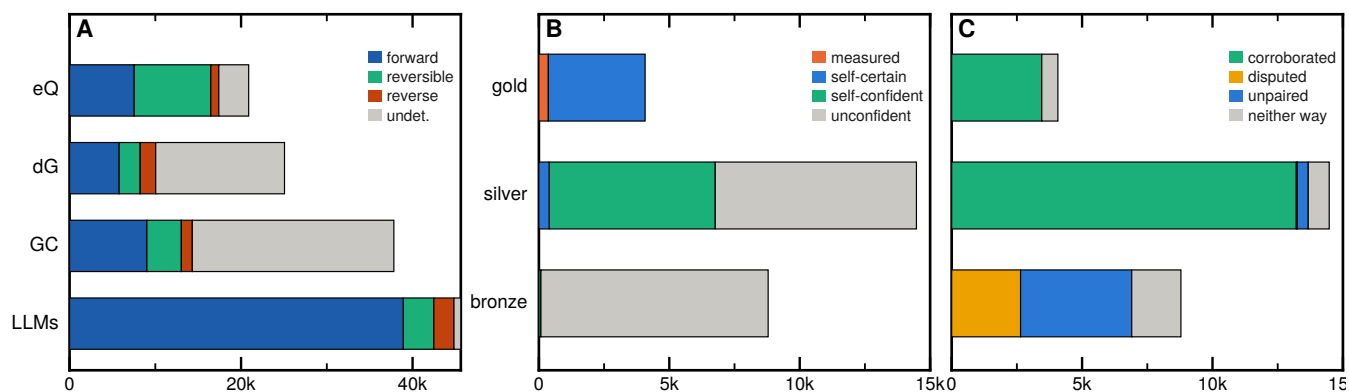


Figure 3 Direction, agreement and evidence. (A) Direction assigned by each source independently: eQ eQuilibrator, dG dGPredictor, GC group contribution, and LLMs the large language model ensemble, which supplies a direction and no energy; its calls are released with the database and take no part in the evidence grading. Bar length is the number of reactions the source covers, so the rows are directly comparable. (B) What each evidence grade rests on by the deciding source's own calibrated confidence, and (C) by what the other sources made of it. The two panels share a scale but not a palette, so each carries its own key; categories are ranked as in Supplementary Table 1. Neither way is the case that table writes ---: a cross-check ran and settled neither way, so it neither helped nor penalised.

eQuilibrator vs dGPredictor

dGPredictor is able to generate a predicted energy for more reactions (~29,600~25,100, against eQuilibrator's ~21,800~17,900) and converts the least of that coverage into a usable reaction direction. For a typical dGPredictor reaction the error bar alone exceeds the decision threshold in the reversibility index, so 58.959.9% of its reactions are returned undetermined against eQuilibrator's 15.72.6%. Where

both sources commit to a direction they agree 95.094.8% of the time. dGPredictor is abstaining rather than contradicting, and it resolves 4,248~3,877 reactions eQuilibrator cannot, against 13,289~11,237 the other way.

Grading.

Of 33,099~27,338 graded reactions, 3,434~4,073 are gold, 18,388 silver and 11,277~14,475 silver and 8,790 bronze (Figure 3DB,C; see classification scheme in Supplementary Methods S3). The

tiers separate on what supports them: gold is ~~7791~~% self-certain and ~~6885~~% corroborated, while bronze is 99% unconfident and split between reactions no second source could check (48%) and reactions the other sources contradict (~~3529~~%). Because a measured reaction is always graded gold, the grades must be recomputed with openTECR withheld before they can be tested against it. Re-graded on the predictors alone, the ~~806 anchored reactions split into 357 gold, 437 silver and 12 801 anchored records split into 439 gold, 349 silver and 13 bronze~~; the gold subset falls within 2 kcal mol^{-1} of measurement ~~96.993.8~~% of the time, against ~~91.389.4~~% for silver. This validation counts records rather than distinct reactions: 438 of the 801 are obsolete duplicates of a live entry, so the effective number of anchored reactions is 363 and these rates rest on fewer independent measurements than the counts imply (Supplementary Methods S3). Both tiers therefore track measurement closely enough that we recommend either for inclusion in a metabolic model, and reserve bronze for review rather than use.

Recommendation.

Evidently where the sources agree on reaction direction, this becomes our recommendation, but because the sources can disagree, we prioritize the source from which we take the recommended reaction direction, so the priority order is eQuilibrator, dGPredictor, group contribution. eQuilibrator supplies ~~21,218–16,985~~ recommended reaction directions, dGPredictor ~~4,248–7,935~~ and group contribution ~~4,6912,055~~; these numbers include the ones where there is agreement also. We record the source used to make the recommendation, so researchers may apply a different precedence.

Atom mapping

Atom mappings are published for ~~32,877 reactions, 5926,242 reactions, 54~~% of the database: ~~25,058–19,809~~ clean, and ~~7,819 6,433~~ that required salvage (which are tagged). As the mappings can be used to span a metabolic reconstruction, reactions needing one repair are likely to carry others.

Reconstruction scope.

A template carries far fewer reactions than the database holds — the v7.0 ModelSEED templates span 8,597 in union, about 9,000 with PlantSEED v3 roles — but that gap does not mean the remainder is unusable. The database is a reference namespace across all kingdoms and includes biochemistry from secondary databases and published models that no single template targets. Promotion into a template is limited by mass and charge balance, which 69% of reactions satisfy, and by association to an annotated functional role, which 41% carry. Gene association is not the constraint: 2,258 of the 8,584 reactions in the Gram-negative template are typed as gapfilling and require no gene. 15,106 reactions now satisfy both balance and role annotation, 8,815 of them outside every current template.

Discussion

Unlike the 2020 release, this update adopts a multi-source policy rather than promoting a single thermodynamic value, publishing four sources, each with their own uncertainty and

estimated direction, because the sources may disagree in ways a single value conceals. Grading against experiment enables us to validate the approach: A researcher can take the highest-graded source and knows what the grade means. Two limits are worth naming. Transport reactions are scored without a membrane potential or a pH gradient, because both are properties of an organism and a condition rather than of a reference reaction; a compartment-aware scoring layer is the natural next release, and it needs the organism context a model supplies and a database cannot. And roughly half the database still carries no thermodynamic direction at all, bounded by the structures needed to compute one. In being explicit about disagreement between sources, we expect this release to serve the widening range of communities that build on shared biochemistry: genome-scale reconstruction, flux analysis, pathway design and the training of predictive models.

Data availability

The ModelSEED Biochemistry Database is available at <https://github.com/ModelSEED/ModelSEEDDatabase> under the Creative Commons Attribution License, as versioned flat files and JSON, and archived at Zenodo (<https://doi.org/10.5281/zenodo.22782602>). Data directly derived from KEGG and MetaCyc remain subject to the licences of those resources. It is also queryable through a Solr endpoint and browsable at <https://modelseed.org>. Released structures, thermodynamic values with per-source uncertainty, evidence grades, atom mappings and the low-confidence reaction flags are all in that repository. The scripts that regenerate every published value, including the eQuilibrator cache rebuild, are under [Scripts/Thermodynamics/](#).

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Author contributions

Contributions follow the CRediT taxonomy. S.M.D.S.: conceptualization, methodology, software, formal analysis, data curation, visualization, writing – original draft, project administration. C.S.H.: conceptualization, funding acquisition, supervision. A.F., J.P.F., V.S. and C.T.: resources, data curation, investigation; J.P.F. also writing – original draft. J.N.E. and F.L.: resources, software (supporting). R.T.G.: the openTECR re-curation of the experimental measurements. E.N. and M.E.B.: eQuilibrator. V.U., M.A. and C.D.M.: dGPredictor. S.H. and Z.N.: atom mapping. A.P.A., R.W.C. and E.M.W.-C.: funding acquisition, supervision. All authors reviewed the manuscript.

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Conflict of interest

None declared.

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